HeatWise

PROJECT GOAL REPORT · 2026

Cooling Cities. One Rooftop at a Time.

A comprehensive goal report covering product vision, strategic objectives, progress metrics, technical architecture, commercial targets, and the full platform roadmap for HeatWise — an AI-powered urban heat reduction platform for Indian cities.

REPORT DATE	STAGE	VERSION
April 8, 2026	Pre-Seed / MVP	v1.0
PLATFORM	**MARKET**	**CATEGORY**
Next.js + Android	India — Tier 1–3	ClimaTech / GreenTech

CONFIDENTIAL

01 · EXECUTIVE SUMMARY

What HeatWise Is Building

HeatWise is an AI-powered platform that reduces urban heat stress in Indian cities by recommending, visualizing, and facilitating the installation of optimized green cover — connecting homeowners and businesses with a verified installer network.

DATABASE MODELS	APP SCREENS	API ENDPOINTS	LIBRARY MODULES
73	27	100+	100+
Prisma schema entities	End-to-end UI flows	Across 19 domains	Domain service files

Mission

Core Mission Statement

To make every rooftop in India a cooling asset — using AI-driven species recommendations, AR-assisted measurement, and a vetted green installer network — so that urban communities can fight extreme heat without waiting for government intervention.

The Problem

Indian cities regularly exceed 45°C in summer. The Urban Heat Island (UHI) effect amplifies ambient temperatures by 3–8°C. Over 90% of Indian rooftops are bare concrete — a massive untapped thermal mass radiating heat into homes and streets. No accessible, AI-guided product exists that takes a homeowner from their specific rooftop conditions to a species plan, installer quote, and verified outcome — until HeatWise.

The Solution

FOR HOMEOWNERS & BUSINESSES

- Scan rooftop via camera and AR measurement
- Receive AI-ranked green cover recommendations
- Visualize before/after thermal transformation
- Connect with and pay verified green installers
- Track CO_2 offset, cooling delta, and biodiversity

FOR THE PLATFORM

- Hybrid ML + rules recommendation engine
- 73-model data layer for telemetry and outcomes
- Installer marketplace with SLA and outcome tracking
- Longitudinal follow-up at 7d / 30d / 90d / 180d
- Full commercial analytics and unit economics

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02 · STRATEGIC GOALS

What We Are Trying to Achieve

Six primary goal pillars drive every product, engineering, and commercial decision at HeatWise. Each goal has a clear scope, success criteria, and dependency chain.

Goal 1 — Deliver a Best-in-Class AI Recommendation Engine

Build a hybrid ML + rules-based recommendation system that produces accurate, reproducible, and explainable green cover plans for any Indian rooftop — regardless of size, budget, climate zone, or soil condition. The engine handles supply constraints, seasonal availability, and budget fit in real time.

ML + RULES HYBRID · TELEMETRY-FIRST · EXPLAINABLE AI

Goal 2 — Build a Verified Installer Marketplace

Create a trust-first installer network covering Tier 1–3 Indian cities. Each installer is capability-verified, SLA-tracked, and commercially integrated. Quotes, job assignments, completion, and outcome tracking all flow through the platform with full auditability.

VERIFIED NETWORK · SLA TRACKING · COMMERCIAL LAYER

Goal 3 — Measure Real-World Impact at Scale

Every completed installation feeds longitudinal outcome data back into the platform. Temperature delta, plant survival rate, CO_2 offset, and user satisfaction are measured at 7d, 30d, 90d, and 180d checkpoints. This data trains the next generation of the ML model.

LONGITUDINAL TRACKING · ML FEEDBACK LOOP · VERIFIED OUTCOMES

Goal 4 — Launch a Production-Ready Android + Web App

Ship a complete 27-screen mobile-first application on Android (Capacitor shell) and web. The app supports phone OTP auth, AR measurement, 3D garden layout visualization, push notifications, and offline-capable project management.

ANDROID + WEB · AR MEASUREMENT · 27 SCREENS

Goal 5 — Achieve Commercial Sustainability

Drive revenue through installer commissions (15% per job), premium PDF reports, and B2B white-label partnerships. Track funnel metrics and unit economics per project to reach contribution-positive economics within 12 months of commercial launch.

15% COMMISSION · B2B WHITE-LABEL · UNIT ECONOMICS

Goal 6 — Operate with Enterprise-Grade Governance

All user data is managed under a full consent and retention policy framework. Telemetry is append-only and immutable. Platform audit trails, idempotency records, and structured error handling ensure the system is compliant, debuggable, and production-scalable from day one.

CONSENT TRACKING · APPEND-ONLY TELEMETRY · AUDIT TRAILS

03 · PRODUCT GOALS

Screen Completion & Sprint Objectives

Current build status of all 27 app screens and active sprint objectives for Q2 2026.

Auth Flow (Login, OTP, Profile)	100%
Splash + Onboarding	100%
Home Dashboard	100%
AR Measure + Photo Capture	90%
Environment Inputs	100%
AI Analysis Screen	100%
Garden Layout (3D Isometric)	85%
Results Screen	100%
Before/After Thermal View	100%
Installer Quote + Install Flow	100%
Species Library + Detail	100%
Impact + Carbon Dashboard	100%
City Heat Map	100%
Notifications / Alerts	70%
Report Screen + PDF Export	85%
Settings + Eco Tips	100%

Sprint Objectives — Q2 2026

OBJECTIVE	DOMAIN	STATUS	TARGET
Real-time push notifications for heat alerts	Frontend	IN PROGRESS	Q2 2026
Flutter UI redesign — Midnight Aurora theme import	Design	PENDING	Q2 2026
3D animated garden layout finalization	Frontend	IN PROGRESS	Q2 2026
Installer SLA dashboard in admin panel	Backend	IN PROGRESS	Q2 2026
Android Capacitor APK release build	Mobile	PENDING	Q3 2026
Species catalog mapping CI verification	Platform	SHIPPED	Done
ML model registry + Python serving	ML	SHIPPED	Done
Recommendation telemetry Phase 8–9	Backend	SHIPPED	Done
Governance and consent framework	Platform	SHIPPED	Done

04 · TECHNICAL GOALS

Architecture & Engineering Objectives

The technical goals underpinning the platform's scalability, accuracy, reproducibility, and operational reliability.

Recommendation Engine

ML LAYER TARGETS

- Python ML service co-deployed with Next.js, scoring species via IPC
- ML registry versioning with per-rollout guardrail checks
- Nightly telemetry pipeline export to training data store
- Candidate snapshot immutability for reproducibility
- A/B experiment assignment with rollout control flags
- Automatic fallback on guardrail failure detection

FALLBACK LAYER TARGETS

- Catalog hybrid fallback covers 100% of species
- Rules-only emergency fallback for ML outages
- Eligibility filtering before candidate scoring
- Supply constraint injection at recommendation time
- Budget fit assessment with downgrade path logic
- Seasonal availability window enforcement

Schema Goals by Domain

DOMAIN	MODELS	PRIMARY GOAL
Recommendation Telemetry	7	Immutable append-only sessions, candidate snapshots, and runtime events
Installer & Commercial	16	Full quote-to-install-to-outcome lifecycle with SLA tracking
Geospatial Intelligence	4	City tier, microclimate, UHI risk, and exposure profile per project
Longitudinal Outcomes	7	Plant survival and temperature tracking at 7d/30d/90d/180d
Governance & Compliance	5	Consent, retention policies, and audit trail enforcement
Supply Chain	5	Species and material availability with regional readiness scores
Reporting & Insights	9	Dossier assembly, PDF export, lesson memory, and segment analytics

Platform Hardening Targets

RELIABILITY

- **Idempotency** — all mutation APIs deduplicated via IdempotencyRecord
- **Structured errors** — machine-readable codes + provenance on every error
- **Audit trail** — PlatformAuditEvent captures all operations append-only
- **Readiness check** — /api/platform/readiness validates DB, ML, and supply before launch

CI / QUALITY GATES

- **TypeScript strict** — tsc --noEmit runs on every push to main
- **Species mapping CI** — generated artifact verified in pipeline
- **Recommendation tests** — integration suite requires seeded DB
- **ML smoke test** — ml:serving-smoke validates Python serving on each deploy

Mobile / Android Target

05 · COMMERCIAL GOALS

Revenue Model & Growth Targets

HeatWise operates a marketplace model. Revenue is generated at the point of successful installation, with secondary streams from premium features and B2B partnerships.

PRIMARY REVENUE
15%
Commission per installer job

SECONDARY REVENUE
B2B
White-label + report exports

PROFITABILITY TARGET
12mo
To positive unit economics

Conversion Funnel Goals

FUNNEL STAGE	EVENT	TARGET RATE	TRACKED VIA
1 · Recommendation	User receives ranked candidates	100% of scans	RecommendationTelemetrySession
2 · Quote Request	User requests installer quote	> 30%	LeadFunnelEvent
3 · Quote Acceptance	User accepts installer quote	> 40% of requests	InstallerQuote
4 · Installation	Job completed and verified	> 85% of acceptances	VerifiedInstallRecord
5 · Follow-Up	30-day outcome data submitted	> 60% of installs	LongitudinalFollowupCheckpoint

Installer Network Goals

COVERAGE TARGETS

- 5 Tier-1 cities at launch — Delhi, Mumbai, Bangalore, Chennai, Hyderabad
- 10 Tier-2 cities by end of Year 1
- Minimum 3 verified installers per city before going live
- Quote response SLA: within 48 hours
- Installation completion SLA: within 14 days of acceptance

QUALITY STANDARDS

- Installer average rating maintained above 4.2 / 5.0
- PartnerSLAMetrics tracked per installer per rolling window
- PartnerCapabilityMatrix verified before onboarding
- Post-install outcome verification mandatory for all jobs
- PartnerFieldOpsStatus updated in real time during active jobs

Unit Economics Targets

METRIC	TARGET	TRACKING MODEL
Average project value	INR 25,000 – 80,000	InstallerQuote
Platform commission per job	INR 3,750 – 12,000	RevenueEvent
Customer Acquisition Cost	< INR 800	CommercialOutcome
LTV / CAC ratio	> 8x by Month 12	CommercialOutcome
Monthly recurring installs	500+ by Month 6	LeadFunnelEvent

06 · ROADMAP

Phase-by-Phase Execution Plan

Six phases from foundation to full commercial scale. Phases 1–3 are complete. Phase 4 is active. Phases 5–6 are planned.

PHASE 1 · COMPLETE · Q4 2025

Core Platform Infrastructure

Next.js setup, Prisma schema (73 models), NextAuth phone OTP auth, recommendation engine skeleton, species catalog seeding, basic API routing across 19 domains. All foundational data models live and tested.

PHASE 2 · COMPLETE · Q1 2026

Recommendation Engine + Telemetry

Hybrid ML + catalog fallback + rules-only fallback. Immutable telemetry sessions and candidate snapshots (Phase 8–9 architecture). Rollout guardrails, A/B assignment, and Python ML serving integration fully operational.

PHASE 3 · COMPLETE · Q1 2026

Full 27-Screen UI — Midnight Aurora Theme

All screens built and integrated. 3D scenes via Three.js, AR measurement overlay, before/after thermal visualization, installer quote flow, impact dashboards, species library, city heat map, and Capacitor Android shell.

PHASE 4 · IN PROGRESS · Q2 2026

Flutter Redesign + Real-Time Alerts

Full Flutter UI system using Midnight Aurora design tokens imported into the project. Real-time push notifications for heat alerts, installer updates, and AI insights. 3D animation upgrade across all 27 screens.

PHASE 5 · PLANNED · Q3 2026

Android Launch + Installer Marketplace Go-Live

Capacitor APK submitted to Google Play Store. Installer marketplace live with real transactions in 5 Tier-1 cities. SLA monitoring active. Longitudinal follow-up scheduler running in production at scale.

PHASE 6 · PLANNED · Q4 2026

Commercial Scale + ML Retraining on Real Data

500+ verified installs per month. ML model retrained on real outcome data from longitudinal tracking. Tier-2 city expansion to 10 cities. B2B white-label product for housing societies and commercial real estate. Unit economics positive.

07 · RISKS & DEPENDENCIES

Risk Register & Mitigation Strategies

Key risks across product, technical, commercial, and market dimensions — each with a defined mitigation path built into the platform architecture.

RISK	CATEGORY	SEVERITY	MITIGATION
ML model underperforms on edge-case Indian climates (coastal, hill stations)	Technical	HIGH	Catalog hybrid and rules-only fallbacks always active. Guardrail failures trigger automatic downgrade with zero user-facing impact.
Installer network too sparse at launch in Tier-2 cities	Commercial	HIGH	Launch restricted to 5 Tier-1 cities. Tier-2 expansion gated on minimum 3 verified installers per city.
AR measurement inaccurate on low-end Android devices	Product	MEDIUM	Manual measurement fallback always available. AR is an enhancement, not a hard requirement for recommendation generation.
Species availability data stale or mismatched to region	Data	MEDIUM	RecommendationConstraintSnapshot captures supply state at recommendation time. Seasonal windows enforced server-side before candidates are scored.
Plant survival rate below prediction in extreme heat zones	Outcome	MEDIUM	LongitudinalFollowupSchedule tracks survival at 4 checkpoints. Outcome data directly feeds the ML retraining pipeline.
User drop-off before completing recommendation flow	Product	MEDIUM	RecommendationTelemetryEvent tracks drop-off at each stage. Phase 4 Flutter redesign targets conversion friction directly.
Recommendation engine latency unacceptable on mobile	Technical	LOW	Python ML subprocess pre-warmed on server start. IdempotencyRecord deduplicates repeat requests. Fingerprinting caches recent responses.
OpenAI API cost spike at high recommendation volume	Commercial	LOW	ML-first architecture reduces GPT dependency to edge cases only. Rules fallback costs zero. Cost-per-recommendation tracked in unit economics model.

Critical Dependencies

EXTERNAL

- **OpenAI API** — GPT-backed analysis and explanation features
- **Runware API** — AI garden visualization image generation
- **SMS Provider** — Phone OTP delivery at login
- **Google Play Store** — Android APK distribution channel
- **Postgres (prod)** — Production DB; replaces SQLite local dev

INTERNAL

- **Python ML service** — Must be co-deployed with Next.js on same server
- **Species catalog artifact** — Must be regenerated and committed after any alias change
- **Installer onboarding** — Commercial launch blocked until 3+ installers per city verified
- **Seeded database** — Recommendation integration tests require seeded DB + DATABASE_URL

08 · IMPACT GOALS

Environmental & Social Targets

HeatWise's north star is measurable impact on urban temperature and quality of life. Revenue and growth are outputs of impact — not the other way around.

COOLING PER INSTALL

-3.5°C

Average surface temp reduction target

CO_2 OFFSET / INSTALL

12kg

Per month per project

YEAR 1 INSTALL TARGET

5,000

Verified rooftop installations

Measurable Impact Metrics

IMPACT METRIC	MEASUREMENT METHOD	YEAR 1 TARGET
Total surface area greened	VerifiedInstallRecord.areaSqm aggregate	500,000 sqm
Average surface cooling per installation	LongitudinalRemeasurement temperature delta avg	-3.5°C
Total CO_2 offset across all installs	CarbonDashboardScreen — platform aggregate	60,000 kg/month
Plant survival rate at 90 days	OutcomeVerificationRecord at 90d checkpoint	> 80%
User satisfaction score	InstallOutcomeRecord.satisfactionScore avg	> 4.3 / 5.0
Cities with active green cover data	GeoContext.cityName distinct count in DB	15 cities
Unique species successfully deployed	VerifiedInstallRecord species variety count	50+ species

Species Cooling Benchmark

Target: Top-10 Cooling Species Deployed in Every City

The SpeciesCatalog ranks plants by cooling score (0–10). HeatWise targets deployment of the top 10 catalog species — Vetiver, Areca Palm, Bamboo, Tulsi, Curry Leaf, Bougainvillea, Moringa, Jasmine, Ficus, and Neem — across all active cities. Vetiver carries the highest cooling score at 9/10 with a measured -4°C surface reduction. The recommendation engine weights these species higher in heat-risk zones where UHI score exceeds 7.

City-Level Contribution Model

HeatWise tracks each installation's fractional contribution to the host city's annual green cover target. The City Heat Map screen surfaces this in real time — showing the aggregate cooling effect of all verified installations across a city. This data becomes the platform's most defensible asset: a ground-truth, installation-by-installation record of urban temperature reduction.

Long-Term Vision — 3 Years

By 2028, HeatWise aims to have contributed to measurable UHI reduction in at least 3 major Indian cities — verifiable via satellite thermal imagery comparison before and after platform deployment. The longitudinal outcome data and city-level aggregation make HeatWise uniquely positioned to demonstrate population-scale climate impact backed by ground-truth installation records — a capability no other platform in the market currently offers.

09 · SUMMARY

Where We Are. Where We Are Going.

A final snapshot of the full platform scope, current build state, immediate next steps, and the closing thesis behind HeatWise.

WHAT IS BUILT — TODAY

- 73-model Prisma schema covering all domains end-to-end
- 27-screen mobile-first UI with 3D animations and AR
- Hybrid ML + catalog + rules recommendation engine
- 100+ API endpoints across 19 service domains
- Full installer quote-to-job-to-outcome lifecycle
- Longitudinal follow-up at 7d / 30d / 90d / 180d
- Enterprise governance: consent, retention, audit trail
- Android Capacitor shell with dev setup complete
- CI pipeline: typecheck + species-mapping verification

WHAT IS NEXT

- **Q2 2026 (Phase 4):** Flutter UI import, real-time alerts, 3D animation upgrade
- **Q3 2026 (Phase 5):** Android Play Store launch, installer marketplace live, 5 cities active
- **Q4 2026 (Phase 6):** ML retraining on real outcomes, Tier-2 expansion, B2B white-label
- **2027:** 10,000+ installs, satellite-verifiable city-level cooling data, Series A raise

Platform Scope at a Glance

PRISMA MODELS	APP SCREENS	API ROUTES	LIB MODULES
73	27	100+	100+

COOLING TARGET	YEAR 1 INSTALLS	LAUNCH CITIES	RUNWAY GOAL
-3.5°C	5,000	5	12mo
Per installation	Verified rooftops	Tier-1 India	To positive unit economics

Closing Statement

HeatWise is not a gardening app. It is urban climate infrastructure — built from the ground up for scale, accuracy, and verifiable real-world impact. Every architectural decision, from immutable telemetry to longitudinal outcome tracking, exists to ensure that every claim the platform makes about cooling a city is backed by auditable, ground-truth data. The goal is not to sell green installations. The goal is to make Indian rooftops a measurable weapon against urban heat — and to prove it with numbers.

HeatWise

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